

P3RS-MT: Privacy-Preserving Personalized Recommendation Services in Medical Tourism

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Abstract—To obtain the most suitable recommendation service for medical tourism, patients typically submit medical records to service providers, who perform similarity computations to recommend customized medical tourism options. To preserve data privacy, patients should upload encrypted medical records, enabling service providers to perform similarity computations on ciphertext. However, medical records are typically high-dimensional and sparse, making encrypted similarity computation less efficient and accurate. In this paper, we propose a privacy-preserving personalized medical tourism recommendation service named P3RS-MT. The proposed scheme employs weighted Tanimoto coefficient (WTC) to enable efficient and accurate similarity computations, while adopts somewhat homomorphic encryption (SHE) to safeguard medical records from exposure to service providers. Since SHE does not natively support division operations—required for WTC—we address this limitation using Newton’s method for iterative approximation. Theoretical analysis and experimental evaluations show that P3RS-MT is secure, accurate and efficient.

Index Terms—Medical tourism, recommendation services, privacy-preserving, similarity computation, homomorphic encryption.

I. INTRODUCTION

HEALTHCARE is a fundamental need in daily life of human beings. To emphasize the significance of healthcare for personal well-being, the United Nations has identified it as one of the Sustainable Development Goals (SDGs) [1]. The development of healthcare technologies, combined with advances in communication and transportation, has contributed to the emergence of medical tourism [2]. Medical tourism refers to individuals traveling to other countries to obtain high-quality and cost-effective healthcare services [3]. According to the 2024–2029 Analysis Report on the Current Situation and Future Development Trends of the Medical Tourism Industry published by CRRP Industry Research Institute [4], the global medical tourism market is expected to USD 239.37 billion by 2029, reflecting a compound annual growth rate (CAGR) of 23.03%.

Nowadays, information on medical tourism can be acquired through specialized online portals, such as Health-Tourism.com [5] and PlacidWay [6]. These platforms typically

rely on basic filtering mechanisms—including keyword matching and elementary criteria such as country and treatment type—to assist users in identifying suitable options. Unfortunately, the results generated by such filtering mechanisms are often too broad and lack specificity. As a result, patients often struggle to identify the most suitable options when confronted with an overwhelming amount of information.

Recommendation systems can help users identify relevant information from massive information. They have been widely adopted across various domains, such as movie recommendation [7], travel information recommendation [8], hotel recommendation [9] and healthcare recommendation [10].

In recent years, deep learning has been extensively adopted in recommendation systems [11], [12]. However, deep learning-based approaches often require large volumes of data and complex model architectures, resulting in high computational cost and prolonged training time. In contrast, traditional recommendation algorithms, including collaborative filtering and content-based filtering, have a simple algorithmic structure and exhibit low computational complexity. Nevertheless, collaborative filtering depends on historical user behavior data, making it challenging to generate effective recommendations for new users or items with limited interaction records, which is a commonly issue referred to as the cold-start problem [13]. Alternatively, content-based recommendation systems extract textual features from user-submitted information to construct measurable vector representations, which are subsequently used to perform similarity computations and generate personalized recommendation lists based on semantic similarity results [14]. By reducing dependency on user-user or item-item correlations, content-based methods can alleviate the cold-start challenge.

Despite this advantage, content-based methods still typically require users to submit personal information to generate personalized recommendations. This reliance on sensitive data introduces privacy concerns, particularly as service providers are generally considered semi-trusted and may potentially use the collected information to analyze or infer additional sensitive attributes. To preserve user privacy, personal information should be encrypted before submission. Consequently, recommendation systems are required to perform all similarity computations directly on ciphertext.

However, similarity computations on ciphertext remain a non-trivial challenge. Bai et al. [15] developed the CryptoMask system based on homomorphic encryption and secure multi-party computation, which supports similarity matching of facial images on ciphertext to protect data privacy in face recognition

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applications. Nevertheless, this system is limited to integer arithmetic and suffers from reduced matching accuracy. Moreover, secure multi-party computation introduces additional communication cost. Yu et al. [16] proposed a trajectory similarity computation protocol under ciphertext domain. This protocol integrates two complementary methods: one based on Euclidean distance to coarsely compare the destination paths of drivers and users, and another employing the Longest Common Subsequence (LCSS) to securely verify the compliance of actual driving trajectories. A notable advantage of this protocol is its support for real number arithmetic. Additionally, the protocol involves a third party, avoiding the communication cost associated with secure multi-party computation.

Nonetheless, protocol [16] is designed for low-dimensional continuous data, such as trajectory points, and is therefore ill-suited for medical tourism records, which are typically high-dimensional, sparse, and often contain noise in the form of outliers and missing values. In particular, the Euclidean distance-based similarity computation used in [16] depends on differences across all dimensions, making it highly sensitive to such noise. Meanwhile, the LCSS-based similarity measure relies exclusively on the length of matching subsequences, without fully capturing the overall numerical distribution. These limitations become especially pronounced in high-dimensional sparse settings, where noise interference and imbalanced data distribution can lead to substantial fluctuations in similarity computation.

To address issues related to high-dimensional data, Bai et al. [15] employed Cosine similarity (CS) in the CryptoMask system. In contrast to the sensitivity to magnitude in Euclidean distance, CS focuses exclusively on the direction of vectors rather than their magnitudes, making it particularly suitable for high-dimensional and sparse datasets [17]. However, when most components of two vectors are zero, their directional information becomes unstable [18], potentially leading to inflated similarity results. This bias can result in false positives and undermine overall accuracy. Such a scenario is highly relevant in medical tourism contexts: since patients undergo different sets of examinations—often limited to specific tests for rare conditions—many medical indicators may remain unrecorded, resulting in sparse vectors with numerous zero entries. Consequently, CS is not well-suited for this application.

In contrast, Zhang et al. [19] constructed a fuzzy keyword search cloud system utilizing Tanimoto coefficient (TC)—an extension of Jaccard similarity (JS) [20] for non-binary data—which offers a more suitable alternative. By considering not only the overlap of identical features but also the contribution of nonidentical features, TC effectively captures similarity in high-dimensional and inherently sparse datasets, as commonly encountered in medical tourism scenarios. Moreover, its insensitivity to zero vectors ensures that it stably reflects the similarity relationship between vectors, even for extremely sparse data.

Building on the foundation of Tanimoto coefficient, it is important to recognize that not all features in medical records contribute equally to clinical decision-making. Assigning differential weights [21] enables critical attributes—such as blood glucose and blood pressure—to exert greater influence

on the similarity measure, while less significant variables play a correspondingly smaller role. This weighting mechanism not only mitigates the effect of outliers but also enhances the stability of similarity computation. Therefore, this paper employs weighted Tanimoto coefficient (WTC) rather than the standard TC, as WTC more faithfully captures patient similarity by accounting for heterogeneous feature importance. Nevertheless, since WTC involves division operations incompatible with somewhat homomorphic encryption (SHE), developing an effective privacy-preserving recommendation service over high-dimensional sparse data remains a critical challenge.

In this paper, we propose a privacy-preserving personalized medical tourism recommendation service, P3RS-MT. It allows service providers to generate recommendation services directly over patients' encrypted medical records, thereby ensuring both data confidentiality and recommendation utility. The main contributions of this paper can be summarized as follows:

- We propose P3RS-MT, a privacy-preserving personalized medical tourism recommendation service. By leveraging SHE, P3RS-MT protects users' private information from exposure to service providers. Furthermore, it incorporates WTC to achieve accurate recommendations while effectively handling high-dimensional and sparse medical records.
- We design an efficient method based on Newton iteration to convert WTC's division operations into multiplicative operations on ciphertext, thereby ensuring compatibility with SHE. This optimization enables efficient encrypted computation, resulting in lower computational cost with higher accuracy compared to CS.
- We conduct comprehensive security and performance analyses to demonstrate that P3RS-MT achieves both privacy preservation and practical efficiency, while maintaining high recommendation accuracy in medical tourism scenarios.

II. RELATED WORK

Existing privacy-preserving techniques for computing vector similarity can be broadly categorized into three paradigms: masking-based lightweight protocols, Secure Multi-Party Computation (SMPC), and Homomorphic Encryption (HE)-based schemes.

Lightweight protocols often utilize random masking or matrix obfuscation. For instance, Xiong et al. [22] proposed a Privacy-Preserving Cosine Similarity Computation (PCSC) protocol for mobile crowd-sensing, employing random masking to obfuscate task vectors. While computationally efficient, such methods typically sacrifice accuracy for performance and lack the rigorous semantic security proofs offered by cryptographic primitives. Furthermore, they are tailored for dense vectors and simple cosine metrics, making them ill-suited for the high-dimensional, sparse data characteristic of medical records. Alternatively, SMPC-based approaches distribute computation among multiple parties to ensure data privacy. Christensen et al. [23] presented an SMPC protocol for computing correlations

using Beaver triples and secret sharing. Although precise, SMPC inherently requires frequent interaction between parties. In a cross-domain medical tourism scenario, strictly synchronous interactions and high communication latency are significant bottlenecks, rendering standard SMPC solutions less scalable for real-time recommendations.

Homomorphic Encryption (HE) enables direct ciphertext computation, categorized by operational capacity [24] into: Partially Homomorphic Encryption (PHE): supports singular operation types, Somewhat Homomorphic Encryption (SHE): supports both addition and multiplication with limited depth, and Fully Homomorphic Encryption (FHE): supports unbounded operations. Some privacy-preserving similarity computations often utilized PHE schemes. For instance, schemes such as [25] and [26] leverage ElGamal or Paillier cryptosystems, while Wang et al. [27] proposed PPSKSQ scheme using symmetric encryption techniques. However, the fundamental limitation of PHE—supporting only a single type of algebraic operation—renders it insufficient for complex similarity computation required in medical records analysis, which typically necessitate both multiplication and addition. Furthermore, schemes relying on symmetric primitives often involve complex key management assumptions that are impractical for open, multi-user medical environments. Keying to the other extreme, FHE schemes, while offering full computational generality, impose substantial computational and communication overhead, making them currently impractical for resource-constrained cloud infrastructures and thus seldom considered for large-scale, latency-sensitive applications like real-time recommendation services.

To balance efficiency and functionality, SHE schemes have emerged as the optimal choice. Prominent SHE instantiations, such as BFV [15] and CKKS [16], natively support both additive and multiplicative homomorphisms. Crucially, they incorporate Single Instruction Multiple Data (SIMD) encoding, which allows multiple plaintext values to be packed into a single ciphertext slot. This enables parallelized arithmetic operations, significantly mitigating the complexity of processing high-dimensional vectors. Despite these advantages, existing SHE-based applications exhibit limitations when applied to medical tourism. The system developed by Bai et al. [15] is based on BFV, which operates strictly on integers. This quantization is ill-suited for medical similarity analysis, where integer-only arithmetic can lead to significant precision loss and degraded matching accuracy. Conversely, the protocol proposed by [16] utilizes CKKS, which effectively addresses this by natively supporting approximate arithmetic on real numbers. However, the specific design in [16] targets low-dimensional, continuous data (specifically, vehicle trajectories) and employs metrics like Euclidean distance and Longest Common Subsequence (LCSS). These metrics are fundamentally incompatible with medical records, which are characterized by high dimensionality and sparsity often containing outliers. In such high-dimensional sparse settings, Euclidean distance is highly sensitive to noise and outliers, while LCSS fails to capture global numerical distributions. Consequently, directly applying the approach in [16] to medical records would yield unstable or inaccurate similarity assessments.

III. PRELIMINARIES

A. Similarity Computation

1) *Tanimoto Coefficient*: Tanimoto coefficient [28] extends the traditional Jaccard coefficient—originally defined by set intersection and union operations—to a continuous form over R^+ , thereby enabling the application in computing similarity between non-Boolean attributes. The similarity computation based on TC coefficient for two nonnegative real vectors $\mathbf{x}, \mathbf{y} \in R^n$ is defined as:

$$\text{Sim}_T(\mathbf{x}, \mathbf{y}) = \frac{\sum_{i=1}^n x_i y_i}{\sum_{i=1}^n x_i^2 + \sum_{i=1}^n y_i^2 - \sum_{i=1}^n x_i y_i} \quad (1)$$

When $\mathbf{x} = \mathbf{y} \neq 0$, $\text{Sim}_T(\mathbf{x}, \mathbf{y}) = 1$; if they are completely uncorrelated or differ significantly in direction, $\text{Sim}_T(\mathbf{x}, \mathbf{y})$ tends to 0.

2) *Weighted Tanimoto Coefficient*: In practical applications, the contribution of individual attributes to similarity measurement typically varies. Weighted Tanimoto coefficient introduced a weight vector $\mathbf{W} = (w_1, w_2, \dots, w_n)$ to reflect the importance differences of each dimensional feature [21], where the generation of $\mathbf{W}$ incorporates TF-IDF logic by integrating Clinical Significance (TF) with Active Frequency (IDF). The similarity computation formula based on WTC for two nonnegative real vectors $\mathbf{x}, \mathbf{y} \in R^n$ is defined as follows:

$$\text{Sim}_{TW}(\mathbf{x}, \mathbf{y}) = \frac{\sum_{i=1}^n w_i x_i y_i}{\sum_{i=1}^n w_i (x_i^2 + y_i^2 - x_i y_i)} \quad (2)$$

When $\mathbf{x} = \mathbf{y} \neq 0$, $\text{Sim}_{TW}(\mathbf{x}, \mathbf{y}) = 1$; when $\mathbf{x} \neq \mathbf{y}$, $\mathbf{x} \neq 0$, and $\mathbf{y} \neq 0$, $\text{Sim}_{TW}(\mathbf{x}, \mathbf{y}) < 1$; when $\mathbf{x} = 0$ or $\mathbf{y} = 0$, $\text{Sim}_{TW}(\mathbf{x}, \mathbf{y}) = 0$.

B. Somewhat Homomorphic Encryption

1) *Cheon-Kim-Kim-Song*: Cheon-Kim-Kim-Song (CKKS) [29] is a widely used SHE scheme that supports a limited number of additions and multiplications over ciphertexts, whose foundation is the Ring Learning with Errors (RLWE) problem [30]. Unlike other homomorphic encryption schemes, CKKS supports approximate arithmetic over real and complex numbers with predefined precision. CKKS treats the noise introduced during decryption as an approximation error for real numbers.

i) Algebraic Structure of CKKS

The M -th cyclotomic polynomial $\Phi_M(X)$, with degree $N = \phi(M)$, defines the fundamental algebraic structure for CKKS. If N is a power of 2, $M = 2N$ and $\Phi_M(X) = X^N + 1$. The core of CKKS operates in the following rings and spaces: $R = \mathbb{Z}[X]/\Phi_M(X) = \mathbb{Z}[X]/(X^N + 1)$ is the ring of polynomials for the plaintext space; $\mathbb{R}_q = R/qR = \mathbb{Z}_q[X]/(X^N + 1)$ is the residue ring for the ciphertext space; and $\mathbb{H}$ is a subspace of $\mathbb{C}^N$ isomorphic to $\mathbb{C}^{N/2}$.

ii) Procedures of CKKS

KeyGen(N): Generate a secret polynomial $s(X)$ and a public polynomial $p(X)$.

Encode(z): Given a message vector $z \in \mathbb{C}^{N/2}$, encoded it into a message polynomial $m(X) = \sigma^{-1}(\lfloor \Delta \cdot \pi^{-1}(z) \rfloor) \in R$, where $\pi^{-1} : \mathbb{C}^{N/2} \rightarrow \mathbb{H}$ lifts z to the subspace $\mathbb{H}$, Δ is a scaling

levels the scheme can support. The final prime with 30-bit can be reserved for special operations.

A modulus chain with more intermediate primes supports a greater number of sequential homomorphic multiplications, but increases ciphertext size and computational cost. Using a larger bit-length primes improves computational precision while adding complexity.

Table I shows typical values of polynomial modulus and their corresponding maximum values of ciphertext modulus.

- **scale:** This parameter determines the output precision. The initial scale is typically set to match the bit-length of an intermediate prime in the modulus chain. For example, if the length of intermediate primes in the modulus chain is 25-bit, the initial scale is usually set to 2^{25} . Each homomorphic multiplication increases the scale, and if the scale approaches the ciphertext modulus, it will degrade the decryption precision.
- **slot size:** Slot size refers to the number of data elements that can be processed in parallel. It depends on the polynomial modulus.

C. Newton Iteration

Newton iteration [33] is used to convert single division to multiple multiplication and addition operations, and the procedures of iterating $\frac{1}{\alpha}$ are described as follows:

Given the function $f(x) = \frac{1}{x} - \alpha$, according to Newton iteration formula:

$$x_{t+1} = x_t - \frac{f(x_t)}{f'(x_t)} \quad (3)$$

Substituting the derivative $f'(x_t) = -\frac{1}{x_t^2}$ of $f(x)$ into (3) yields:

$$x_{t+1} = 2x_t - \alpha x_t^2 = x_t \cdot (2 - \alpha \cdot x_t) \quad (4)$$

To compute $\frac{1}{\alpha}$, the value x_t in (4) is updated iteratively starting from an initial approximation x_0 . The number of required iterations depends on the choice of this initial value.

IV. PROBLEM FORMULATION

A. System Model

During the medical tourism recommendation process, P3RS-MT ensures robust privacy preservation for patients while maintaining uninterrupted service availability. P3RS-MT involves four entities: Data Owners, Medical Tourism Facilitator(MTF), Cryptographic Service Provider(CSP), and User. The main interactions among entities are illustrated in Fig. 1.

- **Data Owners:** Data Owners encompass medical institutions offering tourist medical services as well as historical individual users. Both parties locally extract features from their anonymized medical records to generate structured feature vectors, which were encrypted and then uploaded to MTF as samples.
- **MTF:** MTF is a service provider with powerful computing and storage capabilities. It is responsible for performing similarity computations between encrypted medical

records and samples to generate personalized medical tourism recommendation services for users.

- **CSP:** CSP is an independent cryptographic service provider responsible for key generation, key management, and decryption. Its primary purpose is to secure cryptographic keys.
- **User:** To obtain personalized medical tourism recommendation, users extract and encrypt feature vectors from their medical records. The encrypted vectors are submitted to MTF, which will return tailored recommendation results.

B. Threat Model

The threat model considered in P3RS-MT is described as follows:

- Both MTF and CSP are semi-honest. They follow the protocol exactly, but may attempt to learn sensitive information about users.
- Users honestly upload their medical records to obtain secure and personalized medical tourism recommendation services.
- MTF cannot disclose any sensitive user information to CSP, and there is no collusion between them. In practice, MTFs and CSPs are independent entities. For them, the severe legal liabilities and catastrophic reputational damage make collusion practically unviable.

Furthermore, P3RS-MT should protect the privacy of user's medical records and ensure that users are secure against semi-honest adversaries MTF and CSP:

- When MTF obtains the similarity result through similarity computation, it adds a random number to blind the similarity result. CSP can observe all blinded values received from MTF to infer user's private medical records.
- In a more adversarial setting, MTF may deliberately construct plaintexts and corresponding ciphertexts to infer user's private medical records or other sensitive information. For example, MTF can construct and encrypt several medical feature vectors, and then infer whether the vectors uploaded by users have been involved in similarity computations with them, and whether the constructed vectors are matching samples. This type of attack is called a chosen plaintext attack (CPA) [34].

C. Design Goals and Challenges

1) **Design Goals:** To protect the privacy of user medical records while achieving efficient and accurate personalized recommendation services, P3RS-MT needs to satisfy the following goals.

- **Accuracy:** P3RS-MT should provide accurate similarity results. Firstly, medical records may exhibit uneven feature distributions or excessive sparsity, which can lead to incorrect similarity judgments. Therefore, P3RS-MT should be able to handle sparse data in medical scenarios. Secondly, the similarity computation over ciphertexts is expected to be consistent with those obtained by the same method applied directly over plaintexts.

- *Efficiency*: P3RS-MT should provide efficient recommendation services by minimizing user computational and communication costs. The system performs ciphertext similarity computations on the server-side, while users only perform local data processing, encrypted data uploads, and decryption of encrypted recommendations from MTF.
- *Privacy*: P3RS-MT should protect the privacy of users' medical records. The samples or feature vectors uploaded by users are kept encrypted throughout the entire process to prevent MTF and CSP from inferring the original data. Regarding the final output, MTF is restricted to accessing only the recommendation results necessary for service delivery, whereas CSP remains completely oblivious to these recommendation results.

2) *Design Challenges*: To achieve the aforementioned design goals, the following questions should be considered and addressed by special designs.

Q1: How to achieve accurate, privacy-preserving similarity computation for sparse medical records given that HE explicitly excludes division? High-dimensional, sparse medical records require robust similarity computation involving division, which creates a fundamental conflict with HE's limitation to addition and multiplication. Since standard approximations like Taylor expansion often lack precision or efficiency, the core challenge lies in reconciling these constraints. We must design a privacy-preserving scheme that integrates high-precision iterative numerical methods to approximate division directly on ciphertext, ensuring accurate analysis within HE's computational bounds.

Q2: How to achieve efficient parallel computation for high-dimensional vectors within HE? Processing high-dimensional medical vectors often incurs linear complexity, which is prohibitively slow given the inherent overhead of HE. To meet efficiency requirements, we must move beyond element-wise processing. The key challenge is to design a batch-processing scheme using SIMD capabilities to encode multiple dimensions into single ciphertexts. This allows for parallel arithmetic operations, reducing computational complexity from linear to logarithmic scale and significantly lowering communication costs.

Q3: How to prevent privacy leakage during the collaborative computation between semi-trusted entities? The collaboration between MTF and CSP creates privacy loopholes, enabling MTF to deduce private inputs from query distributions or CSP to access intermediate values. To mitigate this, we must design a secure protocol that grants MTF access strictly to the final recommendation. This mechanism must rigorously prevent both parties from inferring original user vectors or unauthorized intermediate information throughout the interaction process.

V. P3RS-MT SCHEME

A. Design Overview

1) *System Overview and Workflow*: Table II lists the notations and corresponding descriptions used in this paper. P3RS-MT provides privacy-preserving personalized medical tourism recommendation services, and consists of four phases, namely,

TABLE II
NOTATIONS OF P3RS-MT

Notation	Description
n	Degree of polynomial modulus in CKKS
$\epsilon_{Tanimoto}$	Threshold for data similarity
ι	Bit length of the blinded value
κ	Bit length of the domain value
l	Actual length of the user's feature vector
$\boxplus$	Homomorphic addition over ciphertexts
$\boxtimes$	Homomorphic multiplication over ciphertexts
$\hat{\cdot}$	Ciphertext of a single character
$\llbracket \cdot \rrbracket$	Ciphertext of multi characters

System Initialization, Data Encryption and Uploading, Similarity Computation, and Personalized Recommendation Service.

Firstly, on the user side, medical records are transformed into structured feature vectors through local feature extraction, and then encrypted and uploaded to MTF in **Data Encryption and Uploading**. Subsequently, MTF computes the encrypted similarity result between users' records and samples' records using WTC in **Similarity Computation**. Finally, MTF decrypts the similarity result with the assistance of CSP, and determines whether to generate medical tourism recommendation services in **Personalized Recommendation Service**.

2) *Design Rationale*: To address the challenges outlined in Section IV-C, this section provides the technical rationale behind P3RS-MT, elaborating on the specific design logic used to ensure accuracy, privacy, and efficiency.

Integration of WTC with Newton Iteration: We selected WTC for its robustness with sparse data and its inherent support for feature weighting. However, CKKS does not support the division required by WTC. To address this, we integrate Newton iteration to approximate the inverse of the denominator, effectively transforming the division into a sequence of multiplications and additions. This approach is chosen for its quadratic convergence, which ensures high precision within very few iterations. Since the recurrence relation in (4) relies solely on homomorphic multiplication and subtraction—operations natively supported by CKKS—the iteration can be executed directly on the ciphertexts, enabling a seamless and efficient integration with the encryption scheme.

Batch Processing Organization of CKKS: To efficiently handle high-dimensional vectors, we fully leverage the SIMD capability of CKKS. Multiple dimensions of a feature vector are encoded into different "slots" of the same ciphertext. Through cyclic rotations and slot-wise accumulation, we can compute all necessary components for WTC—such as dot products and squared sums—in parallel within a single homomorphic operation. This batch-processing approach significantly reduces the computational complexity.

Secure Collaborative Decryption via Blinding: We implement a blinding-based interaction protocol to resolve the decryption dependency without compromising privacy. Specifically, MTF masks computed ciphertexts with random noise prior to transmission, ensuring CSP observes only meaningless data during decryption assistance. MTF then removes the noise from the returned value to retrieve the final result. This design enforces

Algorithm 1: Encrypt.

Input: Feature vector $\mathbf{H}$ of the user data, and the public key pk_c of CSP.

Output: Encrypted feature vector $\widehat{\mathbf{H}}$ of the user data.

- 1: Calculate $\mathbf{z} \leftarrow \text{toComplex}(\mathbf{H})$;
 - 2: Calculate $u \leftarrow \pi^{-1}(\mathbf{z})$;
 //u: intermediate encode variable
 - 3: Calculate $u \leftarrow u \cdot \Delta$;
 - 4: Calculate $m \leftarrow \sigma^{-1}(u)$;
 - 5: Calculate $c_0 \leftarrow (m - a \cdot pk_c + e)$;
 - 6: Calculate $c_1 \leftarrow a$;
 - 7: Set $\widehat{\mathbf{H}} \leftarrow (c_0, c_1)$;
 - 8: **return** $\widehat{\mathbf{H}}$;
-

a strict separation of privileges, ensuring that CSP provides the decryption capability without accessing sensitive information, while MTF obtains the final result without accessing the decryption key. Throughout this interaction, the user's original feature vectors remain encrypted.

B. System Initialization

In this phase, MTF initializes the system, then CSP and users generate their public-private keys.

- MTF presets and publishes a group of system parameters $(n, \epsilon_{\text{Tanimoto}}, \iota, \kappa)$.
- CSP generates a pair of public-private keys (pk_c, sk_c) under CKKS, publishes pk_c and keeps sk_c secret.
- Each user generates a pair of public-private keys (pk_u, sk_u) using the RSA encryption algorithm [35], publishes pk_u and keeps sk_u secret.
- Data owners pre-encrypt the sample feature vectors using CKKS with CSP's public key and upload them to MTF for storage.

C. Data Encryption and Uploading

In this phase, the user extracts a structured feature vector from medical records, encrypts it using CKKS, and uploads the ciphertext to MTF, as described in Algorithm 1.

The user's medical records are denoted as $D = \{D_T, D_M\}$, where D_T contains textual and numerical data, and D_M contains medical images. The user applies a feature extraction method to transform D into a structured feature vector $\mathbf{H} = (h_1, h_2, \dots, h_l) \in R^l$, where l is the dimension of $\mathbf{H}$. To preserve privacy, the user encrypts $\mathbf{H}$ using CKKS, which operates over complex-valued polynomials.

The real-valued vector $\mathbf{H}$ is first expanded into a complex-valued vector $\mathbf{z} \in \mathbb{C}^{N/2}$ (Line 1) using a mapping function $\text{toComplex}()$, which embeds real numbers into complex slots, typically via interleaving or zero-padding. In our scheme, we set the polynomial modulus degree $N = 16384$ and the scaling factor $\Delta = 2^{30}$ to maintain numerical precision during homomorphic operations.

Subsequently, the encryption process follows the CKKS encode and encrypt procedures outlined in Section III-B1. The

Algorithm 2: Secure Weighted Tanimoto Coefficient Computing (SWTSC) Protocol.

Input: Encrypted feature vector $\widehat{\mathbf{H}}$ of the user data and the encrypted feature vector $\widehat{\mathbf{F}}$ of the sample data.

Output: Plaintext similarity result with noise

$\text{Sim}'_{TW}(\mathbf{H}, \mathbf{F})$

▷ **MTF:**

- 1: Calculate $\widehat{V}_1 \leftarrow \widehat{\mathbf{H}}_i \boxtimes \widehat{\mathbf{F}}_i \boxtimes \mathbf{W}'_i$;
 - 2: Initialize $\widehat{P}_W \leftarrow \widehat{V}_1$;
 - 3: **for** $i = -1, -2, \dots, 1 - n$ **do**
 - 4: $\widehat{P}_W \leftarrow \widehat{P}_W \boxplus \text{Rot}_{\rho_i}(\widehat{V}_1)$;
 - 5: **end for**
 - 6: Calculate $\widehat{V}_2 \leftarrow \widehat{\mathbf{H}}_i \boxtimes \widehat{\mathbf{H}}_i \boxtimes \mathbf{W}'_i$;
 - 7: Initialize $\widehat{S}_H \leftarrow \widehat{V}_2$;
 - 8: **for** $i = -1, -2, \dots, 1 - n$ **do**
 - 9: $\widehat{S}_H \leftarrow \widehat{S}_H \boxplus \text{Rot}_{\rho_i}(\widehat{V}_2)$;
 - 10: **end for**
 - 11: Calculate $\widehat{V}_3 \leftarrow \widehat{\mathbf{F}}_i \boxtimes \widehat{\mathbf{F}}_i \boxtimes \mathbf{W}'_i$;
 - 12: Initialize $\widehat{S}_F \leftarrow \widehat{V}_3$;
 - 13: **for** $i = -1, -2, \dots, 1 - n$ **do**
 - 14: $\widehat{S}_F \leftarrow \widehat{S}_F \boxplus \text{Rot}_{\rho_i}(\widehat{V}_3)$;
 - 15: **end for**
 - 16: Calculate $\widehat{D}_W \leftarrow \widehat{S}_F \boxplus \widehat{S}_H \boxplus \widehat{P}_W$;
 - 17: Calculate $\widehat{R}_W \leftarrow \text{NewtonInverse}(\widehat{D}_W, d)$;
 - 18: Calculate $\llbracket \text{Sim}_{TW}(\mathbf{H}, \mathbf{F}) \rrbracket \leftarrow \widehat{P}_W \boxtimes \widehat{R}_W$;
 - 19: Calculate $\llbracket \text{Sim}'_{TW}(\mathbf{H}, \mathbf{F}) \rrbracket \leftarrow \llbracket \text{Sim}_{TW}(\mathbf{H}, \mathbf{F}) \rrbracket \boxplus < r \mid 0 \mid \dots \mid 0 >$;
 - 20: Send $\llbracket \text{Sim}'_{TW}(\mathbf{H}, \mathbf{F}) \rrbracket$ to CSP.
 - ▷ **CSP:**
 - 21: $\text{Sim}'_{TW}(\mathbf{H}, \mathbf{F}) \leftarrow \text{Decrypt}(\llbracket \text{Sim}'_{TW}(\mathbf{H}, \mathbf{F}) \rrbracket, sk_c)$;
 - 22: Send $\text{Sim}'_{TW}(\mathbf{H}, \mathbf{F})$ to MTF.
-

vector $\mathbf{z}$ is first encoded into a message polynomial $m(X) = \sigma^{-1} \lfloor \Delta \cdot \pi^{-1}(\mathbf{z}) \rfloor$ (Lines 2-4). Then, The polynomial $z(X)$ is encrypted using CSP's public key pk_c to obtain the ciphertext $\widehat{\mathbf{H}} = \text{Enc}(m(X), pk_c)$ (Lines 5-8).

Eventually, the user uploads the ciphertext $\widehat{\mathbf{H}}$ to MTF. At this point, MTF cannot learn the actual length l of the user feature vector, since it is masked by the number of batch slots.

D. Similarity Computation

In this phase, MTF and CSP collaboratively compute the similarity between encrypted user feature vectors and sample feature vectors stored on MTF.

Based on statistical analysis of a multi-domain medical expert knowledge base, MTF generates a weight vector $\mathbf{W}$ to assign importance to each feature dimension. Since $\mathbf{W}$ contains no user's privacy information, it is stored as plaintext. During **System Initialization**, MTF pre-encodes $\mathbf{W}$ into $\mathbf{W}'$ according to CKKS requirements. The process of performing ciphertext similarity computation is detailed in Algorithm 2.

Step 1: After receiving the encrypted feature vector $\widehat{\mathbf{H}}$ from the user, MTF uses SIMD homomorphism with rotation to compute the sum of all weighted element-wise products with

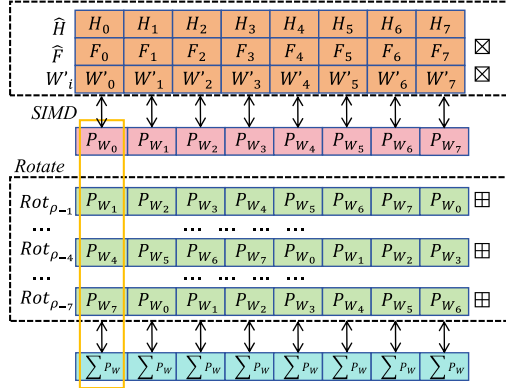


Fig. 2. An example of ciphertexts computation over WTC by using SIMD and rotation, with the number of batched slots is 8.

the encrypted sample vector $\hat{F}$: $\hat{P}_W = \sum_{i=0}^{n-1} \mathbf{W}'_i \boxtimes \hat{H}_i \boxtimes \hat{F}_i$ (Lines 1-3).

Step 2: MTF performs $n-1$ ciphertext rotations on the slot-wise products $\langle \mathbf{W}'_i \cdot \mathbf{H}_i \cdot \mathbf{F}_i \rangle_{i=0}^{n-1}$ within $\hat{P}_W = \sum_{i=0}^{n-1} \mathbf{W}'_i \boxtimes \hat{H}_i \boxtimes \hat{F}_i = \llbracket \sum_{i=0}^{n-1} \mathbf{W}'_i \cdot \mathbf{H}_i \cdot \mathbf{F}_i \rrbracket$ as shown in (5) at the bottom of this page, and then carries out $n-1$ homomorphic additions to aggregate the results (Lines 4-6). As illustrated in Fig. 2, assuming $n=8$, MTF first employs SIMD to perform homomorphic multiplications on $\mathbf{W}'_i$, $\hat{H}_i$ and $\hat{F}_i$. This results in a single ciphertext containing the slot-wise values $[P_{W_0}, \dots, P_{W_7}]$. Subsequently, MTF executes 7 cyclic left-rotations on this ciphertext and aggregates these rotated values via homomorphic addition. Ultimately, this process ensures that every slot in the resulting ciphertext contains the identical accumulated value $\hat{P}_W$ (as discussed in *Integration of WTC with Newton Iteration* in Section V-A2).

Step 3: MTF performs $n-1$ ciphertext rotations on $\langle \mathbf{W}'_i \cdot \mathbf{H}_i \cdot \mathbf{H}_i \rangle_{i=0}^{n-1}$ and $\langle \mathbf{W}_i \cdot \mathbf{F}_i \cdot \mathbf{F}_i \rangle_{i=0}^{n-1}$ based on WTC formula, and then performs homomorphic addition on the rotated ciphertexts (Lines 7-18) to aggregated intermediate results: $\hat{S}_H = \sum_{i=0}^{n-1} \mathbf{W}_i \boxtimes (\hat{H}_i)^2$ and $\hat{S}_F = \sum_{i=0}^{n-1} \mathbf{W}_i \boxtimes (\hat{F}_i)^2$.

Step 4: MTF proceeds to compute WTC. However, since this computation involves division which is not supported by CKKS, MTF first derives the denominator under encryption as: $\hat{D}_W = \hat{S}_F \boxplus \hat{S}_H \boxplus \hat{P}_W$ (Line 19). At this point, each slot of the resulting ciphertext contains $\hat{D}_W$, so MTF directly takes the value in the first slot of $\hat{D}_W$ for further computation.

Algorithm 3: NewtonInverse.

Input: Denominator $\hat{D}_W$ of the similarity computation formula, and number of iterations $d \in \mathbb{N}^+$.

Output: $\hat{R}_W$.

- 1: Initialize $x_0 \leftarrow 0.3$;
- 2: Calculate $\hat{x}_1 \leftarrow x_0 \cdot (2 - \hat{x} \cdot x_0)$;
- 3: **for** $t \leftarrow 1$ **to** $d - 1$ **do**
- 4: $\hat{x}_{t+1} \leftarrow \hat{x}_t \cdot (2 - \hat{x} \cdot \hat{x}_t)$;
- 5: **end for**
- 6: **return** $\hat{R}_W \leftarrow \hat{x}_d$;

To guarantee the convergence of the Newton iteration (i.e., satisfying $|1 - ax_0| < 1$) and ensure numerical stability across diverse boundary conditions, we implement a comprehensive safeguard mechanism. Specifically, since user and sample vectors are pre-normalized ($H_i, F_i \in [0, 1]$), the similarity denominator $\hat{D}_W$ is analytically bounded by $D_{max} = \sum w_i$. By applying a global scaling factor $S = 1/D_{max}$, we map $\hat{D}_W$ into a stable interval $[0.5, 1]$. This normalization ensures that the fixed initial value $x_0 = 0.3$ consistently satisfies the theoretical convergence criteria ($D_W \cdot x_0 \leq 1$), effectively neutralizing the risk of algorithm divergence across heterogeneous or out-of-distribution datasets. Furthermore, to prevent division-by-zero or numerical instability in extremely sparse scenarios, we implement a minimum safety threshold $\lambda = 10^{-6}$ by computing the denominator $D'_W = \hat{D}_W + \lambda$. This small offset maintains recommendation utility while preventing ciphertext overflow caused by magnitude explosion in near-zero scenarios.

Step 5: Once the denominator $\hat{D}_W$ is obtained, to enable homomorphic computation of WTC under CKKS, MTF applies Newton iteration (as discussed in *Batch Processing Organization of CKKS* in Section V-A2) to approximate the inverse of $\hat{D}_W$ (Line 20) to obtain $\hat{R}_W$, the specific procedure is shown in Algorithm 3. Finally, homomorphic multiplication is used to obtain the encrypted similarity result (Line 21):

$$\llbracket \text{Sim}_{TW}(\mathbf{H}, \mathbf{F}) \rrbracket = \hat{P}_W \boxtimes \hat{R}_W \quad (6)$$

Step 6: To prevent CSP from directly learning the similarity result, MTF adds noise to the encrypted similarity result $\llbracket \text{Sim}_{TW}(\mathbf{H}, \mathbf{F}) \rrbracket$ before sending it to CSP (Lines 22-23). Specifically, MTF selects a random number r from the $(\iota - 1)$ -bit space and generates a blinded ciphertext $\llbracket \text{Sim}'_{TW}(\mathbf{H}, \mathbf{F}) \rrbracket$

$$\begin{aligned}
 & \llbracket \mathbf{W}_0 \cdot \mathbf{H}_0 \cdot \mathbf{F}_0 \mid \dots \mid \mathbf{W}_{n-1} \cdot \mathbf{H}_{n-1} \cdot \mathbf{F}_{n-1} \rrbracket \\
 & \boxplus \text{Rot}_{\rho-1}(\langle \mathbf{W}_i \cdot \mathbf{H}_i \cdot \mathbf{F}_i \rangle_{i=0}^{n-1}) = \llbracket \mathbf{W}_1 \cdot \mathbf{H}_1 \cdot \mathbf{F}_1 \mid \dots \mid \mathbf{W}_0 \cdot \mathbf{H}_0 \cdot \mathbf{F}_0 \rrbracket \\
 & \boxplus \text{Rot}_{\rho-2}(\langle \mathbf{W}_i \cdot \mathbf{H}_i \cdot \mathbf{F}_i \rangle_{i=0}^{n-1}) = \llbracket \mathbf{W}_2 \cdot \mathbf{H}_2 \cdot \mathbf{F}_2 \mid \dots \mid \mathbf{W}_1 \cdot \mathbf{H}_1 \cdot \mathbf{F}_1 \rrbracket \dots \\
 & \boxplus \text{Rot}_{\rho_{1-n}}(\langle \mathbf{W}_i \cdot \mathbf{H}_i \cdot \mathbf{F}_i \rangle_{i=0}^{n-1}) = \llbracket \mathbf{W}_{n-1} \cdot \mathbf{H}_{n-1} \cdot \mathbf{F}_{n-1} \mid \dots \mid \mathbf{W}_{n-2} \cdot \mathbf{H}_{n-2} \cdot \mathbf{F}_{n-2} \rrbracket \\
 & = \sum_{j=0}^{n-1} \text{Rot}_{\rho-j}(\langle \mathbf{W}_i \cdot \mathbf{H}_i \cdot \mathbf{F}_i \rangle_{i=0}^{n-1})
 \end{aligned} \quad (5)$$

Algorithm 4: Decrypt.

Input: Encrypted similarity result $\llbracket \text{Sim}'_{TW}(\mathbf{H}, \mathbf{F}) \rrbracket$, and the private key sk_c of CSP.

Output: $\text{Sim}'_{TW}(\mathbf{H}, \mathbf{F})$.

- 1: Calculate $m' \leftarrow \llbracket \text{Sim}'_{TW}(\mathbf{H}, \mathbf{F}) \rrbracket \cdot sk_c$;
- 2: Calculate $v \leftarrow \sigma(m')$;
 //v: intermediate decode variable
- 3: Calculate $v \leftarrow v \cdot \Delta^{-1}$;
- 4: Calculate $z \leftarrow \pi(v)$
- 5: Set $\text{Sim}'_{TW}(\mathbf{H}, \mathbf{F}) \leftarrow z$;
- 6: **return** $\text{Sim}'_{TW}(\mathbf{H}, \mathbf{F})$;

as follows:

$$\begin{aligned} \llbracket \text{Sim}'_{TW}(\mathbf{H}, \mathbf{F}) \rrbracket &= \llbracket \text{Sim}_{TW}(\mathbf{H}, \mathbf{F}) \rrbracket \boxplus \langle r \mid 0 \mid \dots \mid 0 \rangle > \\ &= \left\llbracket \sum_{i=0}^{n-1} \frac{\mathbf{W}_i \cdot \mathbf{H}_i \cdot \mathbf{F}_i}{\mathbf{W}_i \cdot (\mathbf{H}_i)^2 + \mathbf{W}_i \cdot (\mathbf{F}_i)^2} + \langle r \mid 0 \mid \dots \mid 0 \rangle \right\rrbracket \end{aligned} \quad (7)$$

Step 7: Upon receiving the blinded ciphertext $\llbracket \text{Sim}'_{TW}(\mathbf{H}, \mathbf{F}) \rrbracket$, CSP first decrypts following the principle of *Secure Collaborative Decryption via Blinding* in Section V-A2) it to get the similarity result $\text{Sim}'_{TW}(\mathbf{H}, \mathbf{F})$, as shown in Algorithm 4, which contains the noise r . Then, CSP sends $\text{Sim}'_{TW}(\mathbf{H}, \mathbf{F})$ back to MTF (Lines 24–25).

E. Personalized Recommendation Service

As detailed in Algorithm 5, MTF compares the received similarity result with the predefined threshold to determine whether medical tourism services should be recommended.

Upon receiving $\text{Sim}'_{TW}(\mathbf{H}, \mathbf{F})$, MTF verifies whether $\text{Sim}'_{TW}(\mathbf{H}, \mathbf{F}) \geq \epsilon_{Tanimoto} + r$ is satisfied. If the condition holds:

- MTF provides the user with personalized medical tourism service recommendations. Specifically, MTF encrypts the service content $Content$ with the user's public key pk_u , and sends $\llbracket Content \rrbracket$ to the user (Lines 1-2). The $Content$ generally includes basic information about recommended hospitals, service packages and pricing, itinerary arrangements, and medical team.
- After receiving $\llbracket Content \rrbracket$, the user decrypts it using private key sk_u to obtain personalized medical tourism services (Line 6).

VI. THEORETICAL ANALYSIS

A. Security Analysis

As MTF and CSP are considered as semi-honest adversaries, the security of P3RS-MT is defined based on the notion of adaptive semantic security [36]. We first define a leakage function $\mathcal{L}$, which quantifies a reasonable leakage of MTF and CSP under efficiency considerations. Given a user's medical feature vector $\mathbf{H}$ and a sample dataset $\mathcal{M}$, we define: $\mathcal{L}(\mathbf{H}, \mathcal{M}) = AP(\mathbf{H}, \mathcal{M})$ where $AP(\cdot)$ denotes the access pattern.

Algorithm 5: Personalized Recommendation Service.

Input: Plaintext similarity result with noise $\text{Sim}'_{TW}(\mathbf{H}, \mathbf{F})$.

Output: Recommended content for medical tourism services $Content$.

▷ **MTF:**

- 1: **if** $\text{Sim}'_{TW}(\mathbf{H}, \mathbf{F}) \geq \epsilon_{Tanimoto} + r$ **then**
- 2: $\llbracket Content \rrbracket \leftarrow \text{Enc}(Content, pk_u)$;
- 3: **else**
- 4: **return** $\emptyset$;
- 5: **end if**

▷ **User:**

- 6: $Content \leftarrow \text{Dec}(\llbracket Content \rrbracket, sk_u)$;

Definition 1 (Access Pattern): Given a user's medical feature vector $\mathbf{H}$ and a sample dataset $\mathcal{M} = \{\mathbf{m}_0, \dots, \mathbf{m}_{w-1}\}$, the access pattern is described as:

$$AP(\mathbf{H}, \mathcal{M}) = (\mathcal{I}, \text{Sim}'_{TW}(\mathbf{H}, \mathbf{m}_j)_{j \in \mathcal{I}}, h_{\mathbf{m}_j})_{j \in \mathcal{I}}$$

where $\mathcal{I} \subseteq \{0, 1, \dots, w-1\}$ represents the indices of samples that are involved in the similarity computation; $\text{Sim}'_{TW}(\mathbf{H}, \mathbf{m}_j)$ denotes the set of similarity results between the user's feature vector and each sample's feature vector, and $\{h_{\mathbf{m}_j}\}_{j \in \mathcal{I}}$ indicates whether a given sample is a matched one, where:

$$h_{\mathbf{m}_j} = \begin{cases} 1, & \text{If } \text{Sim}_{TW}(\mathbf{H}, \mathbf{m}_j) \geq \epsilon_{Tanimoto} + r \\ 0, & \text{otherwise.} \end{cases}$$

Following the definition of the leakage function and access pattern, the security definition of P3RS-MT is given:

Definition 2 (Adaptive Semantic Security): As a privacy-preserving medical tourism recommendation system, P3RS-MT is adaptively $\mathcal{L}$ -semantically secure against MTF if for all probabilistic polynomial time (PPT) adversaries $\mathcal{A}$, there exists a PPT simulator $\mathcal{S}$ such that:

$$|Pr[\text{Real}_{\mathcal{A}}(1^\lambda) = 1] - Pr[\text{Ideal}_{\mathcal{A}, \mathcal{S}}(1^\lambda) = 1]| \leq \text{negl}(\lambda)$$

where $\text{negl}(\cdot)$ is an negligible function, $\text{Real}_{\mathcal{A}}(1^\lambda)$ and $\text{Ideal}_{\mathcal{A}, \mathcal{S}}(1^\lambda)$ are probabilistic experiments under a semi-honest adversary $\mathcal{A}$, a challenger $\mathcal{C}$, a simulator $\mathcal{S}$, and a stateful leakage function $\mathcal{L}$.

Real_A(1^λ): First, $\mathcal{C}$ runs **System Initialization** to generate system parameters and the key-pair (pk_c, sk_c) . Then, $\mathcal{A}$ outputs a medical feature vector $\mathbf{H}$ with dimension $l \leq n$, along with an adaptively chosen sample dataset $\mathcal{M} = \{\mathbf{m}_0, \dots, \mathbf{m}_{w-1}\}$. Subsequently, $\mathcal{A}$ and $\mathcal{C}$ run **Similarity Computation** to get the similarity result $\text{Sim}'_{TW}(\mathbf{H}, \mathbf{m}_j)_{j \in \mathcal{I}}$. If there exists some $j \in \mathcal{I}$ such that $\text{Sim}'_{TW}(\mathbf{H}, \mathbf{m}_j) \geq \epsilon_{Tanimoto} + r$, $\mathcal{A}$ and $\mathcal{C}$ run **Personalized Recommendation Service** based on $h_{\mathbf{m}_j} \in \mathcal{I}$. Specifically, for each sample $\mathbf{m}_j$, $\mathcal{C}$ acts as a user and interacts with $\mathcal{A}$ acting as MTF to compute $\text{Sim}'_{TW}(\mathbf{H}, \mathbf{m}_j)$. Finally, $\mathcal{A}$ outputs a bit b as the outcome.

Ideal_{A,S}(1^λ): First, $\mathcal{S}$ runs **System Initialization** to generate system parameters and the key-pair (pk_c, sk_c) . Then, $\mathcal{A}$ outputs a medical feature vector $\mathbf{H}$ with dimension $l \leq n$, along with an adaptively chosen sample dataset $\mathcal{M} = \{\mathbf{m}_0, \dots, \mathbf{m}_{w-1}\}$. Meanwhile, $\mathcal{S}$ is given $\mathcal{L}(\mathbf{H}, \mathcal{M}) =$

$AP(\mathbf{H}, \mathcal{M})$. Afterwards, $\mathcal{A}$ and $\mathcal{S}$ run **Similarity Computation** to compute the similarity result $\text{Sim}'_{TW}(\mathbf{H}, \mathbf{m}_j)_{j \in \mathcal{I}}$. If there exists some $j \in \mathcal{I}$ such that $\text{Sim}'_{TW}(\mathbf{H}, \mathbf{m}_j) \geq \epsilon_{Tanimoto} + r$, $\mathcal{A}$ and $\mathcal{S}$ run **Personalized Recommendation Service** based on $h_{m_j}_{j \in \mathcal{I}}$. Specifically, for each sample $\mathbf{m}_j$, $\mathcal{S}$ acts as a user and interacts with $\mathcal{A}$ acting as MTF to compute $\text{Sim}'_{TW}(\mathbf{H}, \mathbf{m}_j)$. Finally, $\mathcal{A}$ outputs a bit b as the outcome.

Definition 3 (Security of Server-Side Protocol): Suppose there is an protocol between MTF and CSP, where MTF and CSP is required to output functions $F_{MTF}(\llbracket \text{Sim}_{TW}(\mathbf{H}, \mathbf{m}_j) \rrbracket; sk)$ and $F_{CSP}(\llbracket \text{Sim}_{TW}(\mathbf{H}, \mathbf{m}_j) \rrbracket; sk)$ respectively. Let $\text{View}_{MTF}(\llbracket \text{Sim}_{TW}(\mathbf{H}, \mathbf{m}_j) \rrbracket; sk)$ (*resp.* $\text{View}_{CSP}(\llbracket \text{Sim}_{TW}(\mathbf{H}, \mathbf{m}_j) \rrbracket; sk)$) denote MTF's (*resp.* CSP's) view during the execution of the server-side protocol, then there is:

$$\text{View}_{MTF}(\llbracket \text{Sim}_{TW}(\mathbf{H}, \mathbf{m}_j) \rrbracket; sk) = (\llbracket \text{Sim}_{TW}(\mathbf{H}, \mathbf{m}_j) \rrbracket, r_{MTF}, \text{message}_1, \dots, \text{message}_t)$$

where r_{MTF} , r_{CSP} denotes MTF's (*resp.* CSP's) randomness, message_i represent the message exchanged between the two parties. Let $O_{MTF}(\llbracket \text{Sim}_{TW}(\mathbf{H}, \mathbf{m}_j) \rrbracket; sk)$ and $O_{CSP}(\llbracket \text{Sim}_{TW}(\mathbf{H}, \mathbf{m}_j) \rrbracket; sk)$ denote MTF's and CSP's output. We say that the server-side protocol is secure against semi-honest adversaries if there exist PPT simulators $\mathcal{S}_{MTF}$ and $\mathcal{S}_{CSP}$ such that:

$$\begin{aligned} &(\mathcal{S}_{MTF}(\llbracket \text{Sim}_{TW}(\mathbf{H}, \mathbf{m}_j) \rrbracket, F_{MTF}(\llbracket \text{Sim}_{TW}(\mathbf{H}, \mathbf{m}_j) \rrbracket; sk)), \\ &F_{CSP}(\llbracket \text{Sim}_{TW}(\mathbf{H}, \mathbf{m}_j) \rrbracket; sk)) \equiv (\text{View}_{MTF}(\llbracket \text{Sim}_{TW}(\mathbf{H}, \mathbf{m}_j) \rrbracket; sk), \\ &O_{CSP}(\llbracket \text{Sim}_{TW}(\mathbf{H}, \mathbf{m}_j) \rrbracket; sk)), (F_{MTF}(\llbracket \text{Sim}_{TW}(\mathbf{H}, \mathbf{m}_j) \rrbracket; sk), \\ &O_{MTF}(\llbracket \text{Sim}_{TW}(\mathbf{H}, \mathbf{m}_j) \rrbracket; sk)), \\ &\text{View}_{CSP}(\llbracket \text{Sim}_{TW}(\mathbf{H}, \mathbf{m}_j) \rrbracket; sk)) \end{aligned}$$

Following the above security definitions, the security analysis of P3RS-MT is shown below:

Theorem 1: P3RS-MT is adaptively $\mathcal{L}$ -semantically secure against the semi-honest MTF.

Proof: We describe a PPT simulator $\mathcal{S}$ such that the outputs of $\text{Real}_{\mathcal{A}}(1^\lambda)$ and $\text{Ideal}_{\mathcal{A}, \mathcal{S}}(1^\lambda)$ are computationally indistinguishable for any semi-honest PPT adversary $\mathcal{A}$.

First, $\mathcal{S}$ runs **System Initialization** to generate system parameters and the key-pair (pk, sk) . Then, $\mathcal{A}$ outputs a medical feature vector $\mathbf{H}$ with dimension l , along with an adaptively chosen sample dataset $\mathcal{M} = \{\mathbf{m}_0, \dots, \mathbf{m}_{w-1}\}$.

Given the leakage function $\mathcal{L}(\mathbf{H}, \mathcal{M})$, $\mathcal{S}$ randomly generates:

- The medical feature vector $\tilde{\mathbf{H}}$ of the user with dimension $l \leq n$ and the sample feature vector $\tilde{\mathbf{m}}_j$ with $\text{Sim}'_{TW}(\tilde{\mathbf{H}}, \tilde{\mathbf{m}}_j) = \text{Sim}'_{TW}(\mathbf{H}, \mathbf{m}_j)$.
- A sample dataset $\tilde{\mathcal{M}} = \{\tilde{\mathbf{m}}_0, \dots, \tilde{\mathbf{m}}_{w-1}\}$ and $\tilde{h}_{m_j} = h_{m_j}$ for all $j \in [0, w)$.

Subsequently, $\mathcal{A}$ and $\mathcal{S}$ run **Similarity Computation** to obtain $\text{Sim}'_{TW}(\mathbf{H}, \mathbf{m}_j)_{j \in \mathcal{I}}$. If there exists some $j \in \mathcal{I}$ such that $\text{Sim}'_{TW}(\mathbf{H}, \mathbf{m}_j) \geq \epsilon_{Tanimoto} + r$, $\mathcal{A}$ and $\mathcal{S}$ run **Personalized Recommendation Service** over $\mathcal{M}$ and $\mathbf{H}$. In particular, for each sample $\mathbf{m}_j \in \mathcal{M}$, $\mathcal{S}$ acts as a user to interact with $\mathcal{A}$ acting as MTF. $\mathcal{A}$ cannot distinguish $\tilde{\mathbf{H}}$ from $\mathbf{H}$, $\tilde{\mathbf{m}}_j$ from $\mathbf{m}_j$

during **Data Encryption and Uploading**, and cannot distinguish $\tilde{\mathcal{M}} = \{\tilde{\mathbf{m}}_0, \dots, \tilde{\mathbf{m}}_{w-1}\}$ from $\mathcal{M} = \{\mathbf{m}_0, \dots, \mathbf{m}_{w-1}\}$ during **Similarity Computation**, because they are encrypted using CKKS. Furthermore, the length l is masked in the n slots. $\square$

Theorem 2: P3RS-MT is secure against the semi-honest CSP.

Proof: In **Similarity Computation**, MTF requests blindness decryption from CSP, where the actual value is hidden from CSP with statistical security of $2^{\kappa-\ell+1}$. In order to prove this theorem, we construct simulators in the following two distinct scenarios.

Scenario 1: Suppose CSP is compromised by an adversary. We construct a simulator $\mathcal{S}_{CSP}$ to simulate CSP's view View_{CSP} . When MTF blinds the set of ciphertexts $\{\llbracket \text{Sim}'_{TW}(\mathbf{H}, \mathbf{m}_j) \rrbracket\}_{j \in \mathcal{I}}$ into $\{\llbracket \text{Sim}'_{TW}(\mathbf{H}, \mathbf{m}_j) \rrbracket\}_{j \in \mathcal{I}}$ and intends to send them to CSP, let $\mathcal{S}_{CSP}$ randomly select and encrypt a set of $\ell - \text{bit}$ random numbers, matching the size of the $\llbracket \text{Sim}'_{TW}(\mathbf{H}, \mathbf{m}_j) \rrbracket$. Then, $\mathcal{S}_{CSP}$ sends the encrypted random numbers to CSP, which decrypts them upon receipt. Since the blinded ciphertexts are constructed by masking the original values with uniformly distributed random numbers, which are independent of MTF's inputs, no PPT adversary can distinguish between the random numbers and the blinded ciphertexts.

When MTF intends to send the ciphertext of similarity results h_{m_j} to CSP, $\mathcal{S}_{CSP}$ simulates as follows: Randomly selects w integers from $\{0, 1\}$, denoted $h'_m = (h'_{m_0}, \dots, h'_{m_{w-1}})$, computes $\widehat{h'_m} = \text{Enc}(\langle h'_{m_0}, \dots, h'_{m_{w-1}} \rangle, pk)$, and sends $\widehat{h'_m}$ to CSP. After decryption, CSP obtains h'_m which follows the same distribution with h_m , and independent of MTF's inputs, and no PPT adversary can distinguish between h'_m and h_m . Therefore, the distributions of $\mathcal{S}_{CSP}$ and View_{CSP} are indistinguishable.

Scenario 2: Suppose MTF is compromised by an adversary. We construct a simulator $\mathcal{S}_{MTF}$ to simulate MTF's view View_{MTF} . When CSP sends a batch of $\{\text{Sim}'_{TW}(\mathbf{H}, \mathbf{m}_j)_{j \in \mathcal{I}}\}$ to MTF, let the simulator $\mathcal{S}_{CSP}$ encrypt a set of w -bit random numbers and send the ciphertexts to MTF. These ciphertexts cannot be distinguished from $\{\text{Sim}'_{TW}(\mathbf{H}, \mathbf{m}_j)_{j \in \mathcal{I}}\}$ due to the semantic security of CKKS. When CSP wants to send comparison result h_m to MTF, $\mathcal{S}_{MTF}$ randomly selects and generates a batch of w integers in $\{0, 1\}$, and sends them to MTF. Since the generated values follow the same distribution as h_m and independent of CSP's inputs, no PPT adversary can distinguish them from h_m . Therefore, the distributions of $\mathcal{S}_{MTF}$ and View_{MTF} are indistinguishable.

Above two scenarios between MTF and CSP prove that P3RS-MT is secure against the semi-honest CSP according to Definition 3. $\square$

From the above security analysis, it can be concluded that P3RS-MT satisfies the security requirements under the threat model proposed in Section IV-B.

B. Convergence Analysis

Theorem 3 (Robustness of Newton Iteration): Let ξ denote the bounded encryption noise introduced by homomorphic operations in a single iteration step. The relative error E_{n+1} at step $n + 1$ satisfies the recurrence inequality: $|E_{n+1}| \leq E_n^2 +$

$O(\xi)$. The iteration remains convergent as long as the squared approximation error dominates the encryption noise.

Proof: The standard Newton iteration for computing the inverse $\frac{1}{a}$ is defined by the update function $F(x) = x(2 - ax)$. In the CKKS homomorphic encryption scheme, due to the approximate nature of arithmetic operations (such as rescaling and multiplication), the actual update function is modified by an additive noise term ξ , denoted as $\tilde{F}(x) = x(2 - ax) + \xi$. Let $\tilde{x}_n$ be the computed value at step S . We define the relative approximation error at step n as $E_n = 1 - a\tilde{x}_n$. By substituting the noisy update $\tilde{x}_n + 1 = x_n(2 - a\tilde{x}_n) + \xi$ into the error expression for the next step, we obtain: $E_{n+1} = 1 - a[\tilde{x}_n(2 - a\tilde{x}_n) + \xi]$, $E_{n+1} = 1 - (2a\tilde{x}_n - a^2\tilde{x}_n^2) - a\xi$ and $E_{n+1} = (1 - a\tilde{x}_n)^2 - a\xi = E_n^2 - a\xi$. It follows that $|E_{n+1}| \leq |E_n|^2 + |a||\xi|$. This recurrence demonstrates that the error decreases quadratically (E_n^2) during the initial phase. The subsequent stabilization of the error (the Noise Floor) rather than its divergence provides evidence that the iteration has successfully converged to the theoretical limit of the HE scheme. This justifies the use of a limited number of iterations ($d = 3$) to achieve an optimal balance between computational efficiency and numerical stability under noise accumulation. $\square$

VII. COMPLEXITY ANALYSIS

This section will analyze the computational and communication costs for different entities in P3RS-MT. We disregard the costs associated with the offline phase-such as key generation and random number selection-as these can be completed in advance, independently of **similarity computation**. Furthermore, we account for computational costs of encryption, decryption, homomorphic addition, homomorphic multiplication, and ciphertext rotation in CKKS. Communication costs are quantified by evaluating the total number of transmitted ciphertexts.

A. User Side

During **Data Encryption and Uploading**, the user only needs to perform one encryption operation on the preprocessed feature vector. And then transmit one ciphertext to MTF. Therefore, the communication cost consists of transmitting a single ciphertext. During **Personalized Recommendation Service**, if the similarity result meets the predefined threshold, the user receives one ciphertext from MTF and performs one decryption operation to obtain the recommended content.

B. Server Side

The server-side include MTF and CSP. During **Similarity Computation**, the computational complexity of MTF or CSP is analyzed as follows:

Step 2: MTF performs two homomorphic multiplications, $n-1$ ciphertext rotations, and $n-1$ homomorphic additions as shown in (5).

Step 3: MTF performs $2(n-1)$ ciphertext rotations, $2(n-1)$ homomorphic additions, and four homomorphic multiplications as shown in lines 7-17 of Algorithm 2.

Step 4: MTF performs one homomorphic multiplication and one homomorphic subtraction as shown in Line 19 of Algorithm 2.

Step 5: MTF executes three Newton iterations to approximate the reciprocal of the similarity denominator. Each iteration includes two homomorphic multiplications and one homomorphic subtraction, totaling six homomorphic multiplications and three homomorphic subtractions in Algorithm 3. Finally, one additional homomorphic multiplication is required to compute the weighted similarity result based on the approximate reciprocal of the denominator, as shown in (6).

Step 6: MTF performs one homomorphic multiplication to generate the blinded ciphertext as shown in (7), which is then transmitted to CSP.

Step 7: CSP performs one decryption and returns the plaintext result to MTF.

During **Personalized Recommendation Service**, MTF performs one encryption and sends the encrypted recommendation result to the user.

In total, MTF performs $3(n-1)$ ciphertext rotations, $3(n-1)$ homomorphic additions, fifteen homomorphic multiplications, and four homomorphic subtractions. Although the number of rotations and additions is relatively high, these operations in CKKS are applied to ciphertexts encoded in SIMD form, enabling parallel execution over n slots. Moreover, modern homomorphic encryption (HE) libraries optimize rotation operations, substantially reducing actual runtime.

In terms of communication, the entire protocol involves the transmission of two ciphertexts: one from MTF to CSP, and the other from MTF to the user.

VIII. EXPERIMENT EVALUATION

A. System Implementation

Datasets: Two publicly available medical datasets are used for the experiments in this section, and the details for each dataset are presented below:

- Wiscosin Breast Cancer dataset (WBCD): This dataset [37] contains morphological features of cell nuclei (e.g., radius, texture, borders, etc.), which is used to simulate breast cancer diagnostic scenarios. These data points can be extracted using the Active Contour Model (Snakes algorithm) [38].
- Diagnosis Alzheimer With handwriting (DARWIN): The dataset of Alzheimer's Disease (AD) downloaded from UC Irvine [39], which is used to evaluate model performance on high-dimensional sparse medical data. It contains 452-dimensional features for 174 participants (89 AD patients and 85 healthy people). These features can be extracted through kinematic and dynamic signal analysis. [40]

System Configuration: P3RS-MT is implemented in Python using the TenSEAL library [41]. CKKS is adopted, with the polynomial modulus degree $n=16384$. By using SIMD homomorphism, 16384 plaintexts can be efficiently batched into a single ciphertext. The bit-length of the domain value is set to $\kappa=16$, and the bit-length of the blinded value is set to $\iota=56$. All experiments are performed on a computer with an Intel Core i5-13500 processor (12 cores, 16 threads, up to 4.7 GHz) and 16 GB of RAM.

B. Evaluation Metrics

To quantify the quality of recommendation results and identify potential areas for further optimization, several common evaluation metrics are adopted, as detailed follows:

Root Mean Square Error (RMSE) measures the average squared difference between the predicted values and the corresponding ground truth. It is computed as $RMSE = \sqrt{\frac{\sum_{i=1}^n (e_i - r_i)^2}{n}}$, where n is the number of samples, e_i denotes the predicted value, and r_i is the true value.

False Negative Rate (FNR) represents the proportion of similar samples not successfully recognized in ciphertext. It is computed as $FNR = \frac{FN}{TP+FN}$, where True Positives (TP) are records correctly recognized as similar in both plaintext and ciphertext, while False Negatives (FN) are records identified as similar in plaintext but not recognized correctly in ciphertext.

True Positive Rate (Recall) represents the proportion of similar samples successfully recognized. It is computed as $Recall = \frac{TP}{TP+FN}$.

Approximation Error (AE) represents the difference between the results obtained by an approximation method and the true values. It is quantified through the absolute difference and is used to assess the error introduced by approximation. It is computed as $AE = |S_{approx} - S_{true}|$, where S_{approx} denotes the results obtained via the approximation method, and S_{true} is the true value. This error is commonly used to evaluate the precision of encryption computations or other approximate methods.

Top-K Precision (Precision@K) evaluates the proportion of correctly matched samples among the Top-K most similar records. It is computed as $Precision@K(x_q) = \frac{k_{match}}{K}$, where x_q is a given query sample, K is the number of most similar samples (excluding itself), denoted as Top-K. k_{match} is the number of Top-K results that share the same label as x_q . The final $precision@K$ is obtained by averaging over all query samples, namely $Precision@K = \frac{1}{N} \sum_{q=1}^N \frac{k_{match}}{K}$.

Mean Average Precision at K (MAP@K) evaluates the average ranking precision of relevant samples within the Top-K recommendation results. It is computed as $AP@K(x_q) = \frac{1}{m} \sum_{i \in Relevant} P(i)$, where $P(i) = \frac{\# \text{ of relevant samples in top } i}{i}$, x_q is a given query sample, and $[y_1, y_2, \dots, y_k]$ are the labels of its Top-K most similar records. y_q is the true label of x_q . A position $i \in [1, k]$ is counted as relevant if $y_i = y_q$, and m is the number of such positions within Top-K. The final $MAP@K$ is obtained by averaging $AP@K(x_q)$ over all query samples, namely $MAP@K(x_q) = \frac{1}{N} \sum_{q=1}^N AP@K(x_q)$.

C. Experiment Results

We first implement P3RS-MT to evaluate its effectiveness. Each clinical record is processed via Min-Max normalization (mapping all attributes to the unit interval [0,1]), which eliminates the impact of varying magnitudes while strictly satisfying the theoretical convergence domain of the Newton iteration. Subsequently, 100 records are randomly selected from the normalized datasets. To perform the WTC computation, each attribute is assigned a clinical weight w_i based on its importance in diagnosis, which defines the boundary for the scaling factor S

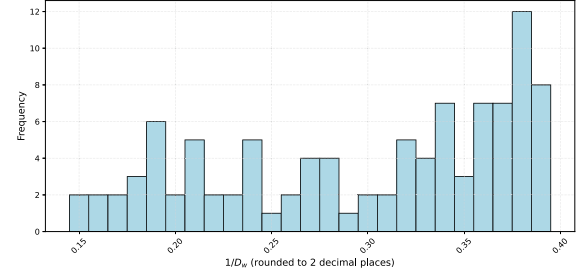


Fig. 3. Frequency distribution of $1/D_W$ values.

to ensure numerical stability during homomorphic operations. Furthermore, DARWIN is used to evaluate the accuracy of different similarity computation methods on sparse data, and these methods include CS, JS, TC, and WTC.

1) Newton Iteration Parameter Settings: P3RS-MT employs Newton iteration to convert division into multiplication within WTC. Since the initial value and iteration rounds affect Newton iteration efficiency, this section empirically determines suitable parameter values. The division within WTC is defined as $\frac{1}{D_W}$ in Line 19 of Algorithm 2.

Firstly, selecting an appropriate initial value x_0 for Newton iteration is crucial, as the closer the initial value approximates the true inverse, the smaller the resulting error, as demonstrated in Section III-C.

The initial value x_0 is computed using a weighted average: $x_0 = \sum_{v \in V} \frac{freq(v)}{N} \cdot v$, where N is the number of samples, V is the value set of computed $\frac{1}{D_W}$ of samples, and $freq(v)$ denotes the frequency of value v among the samples.

Fig. 3 shows the distribution of $\frac{1}{D_W}$, and each value is calculated between the user record and all 100 sample records in the normalized dataset. It can be seen that most values fall within the range [0.15, 0.39].

Secondly, iteration rounds can affect the performance of Newton iteration. To determine this parameter, ten test samples are constructed over the interval [0.1, 1.0] with a step size of 0.1, where the initial value is 1.5. The iterations are carried out for one to six rounds, respectively. Then, we compute RMSE for Newton iteration with all rounds, where e_i denotes the value obtained through Newton iteration, and r_i represents the corresponding true value.

Fig. 4 presents error convergence across iteration rounds, while Fig. 5 shows corresponding RMSE and runtime metrics. From the figures, it can be observe following:

- Accuracy improves with iteration count: Results significantly deviate from true values after one iteration, but errors converge after three iterations.
- Computational cost increases with iterations: Time consumption rises slowly through three iterations, accelerates between iterations three and four, then stabilizes.
- RMSE stabilizes after four iterations: It drops below 0.5 after three iterations with marginal changes thereafter.

Increasing iteration rounds reduces similarity computation errors but incurs higher computational costs. Therefore, three iterations provide an optimal balance between accuracy and efficiency.

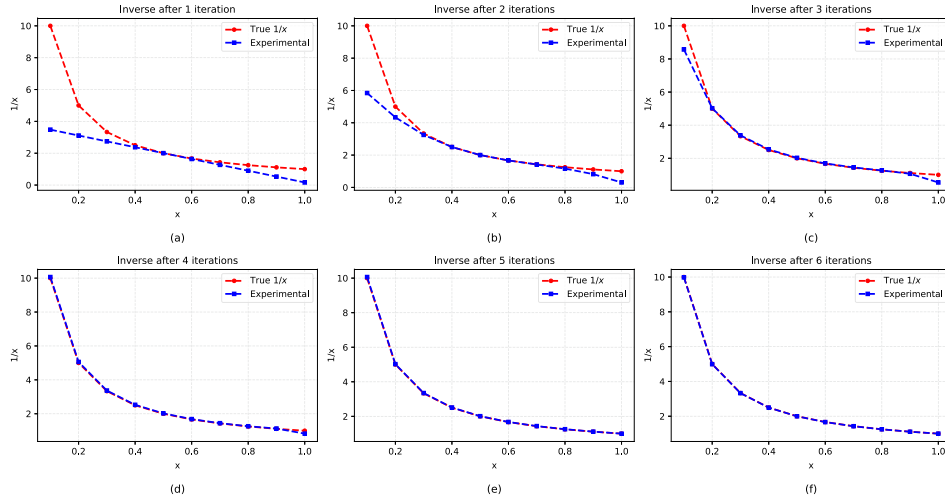


Fig. 4. Approximate inverse values after each iteration. True values are $1/x$ computed in plaintext without error, experimental values are $1/x$ computed using CKKS.

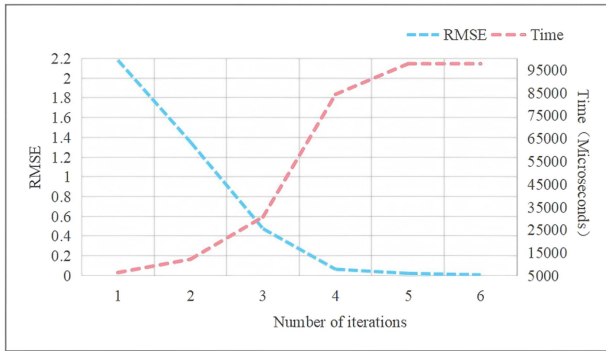


Fig. 5. Inverse RMSE values and running time for 6 iterations.

2) *Performance Evaluation*: i) *Experiments on WBCD*: We conducted experiments on pre-processed WBCD to evaluate the accuracy and efficiency of P3RS-MT, where decisive morphological indicators (radius mean and concavity mean) are assigned higher weights based on their clinical significance. CKKS modulus chain parameters are set as $[40,30,30,30,30,30,30,30,30,30,30,40]$, and the scaling factor as 2^{30} . To assess the sensitivity of P3RS-MT, we introduce a random perturbation (within a $\pm 10\%$ range) to the weight vector $\mathbf{W}$. The consistency of the resulting recommendations is evaluated by the Jaccard similarity index, which qualifies the membership overlap between the baseline Top-10 list and the perturbed list. Empirical results indicate that the Jaccard scores consistently range from 0.82 to 1.00. Although minor rank reorderings occur within the Top-10 lists (e.g., a positional swap between the 5th and 6th candidates), the precise membership of the recommendation set remains highly stable. This confirms that the P3RS-MT system exhibits robust resilience against minor variations in $\mathbf{W}$. Consequently, despite these fluctuations, the core utility of the service—delivering the most relevant medical provider cases to the user—is effectively preserved.

a) *Accuracy and efficiency analysis of P3RS-MT*:

TABLE III
EVALUATION OF RECOGNITION RESULTS UNDER DIFFERENT SIMILARITY THRESHOLDS

	(Initial Value = 0.3)		
$\epsilon_{Tanimoto}$	0.65	0.70	0.75
TP	51	36	25
FN	0	1	1
FNR	0	2.7%	3.85%
Recall	1	97.3%	96.15%

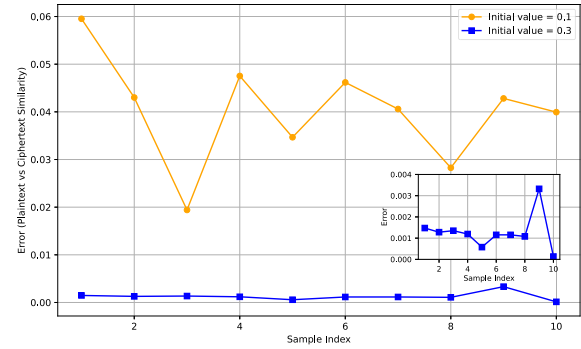


Fig. 6. Approximate error within two initial values under three iterations within 10 tests.

Accuracy: The evaluation first examines FNR and Recall, then computes AE between P3RS-MT similarity results and those obtained from plaintext.

First, we compute the FNR and Recall of P3RS-MT, with results shown in Table III, where the initial value is 0.30 as specified in Section III-C. The results show that increasing the similarity threshold $\epsilon_{Tanimoto}$ leads to higher FNR and lower Recall. When $\epsilon_{Tanimoto} = 0.65$, FNR reaches 0 and Recall achieves 100%, indicating optimal recommendation performance.

Second, we compute AE with initial values of 0.30 and 0.10, where 0.10 is the commonly used choice. The results are shown in Fig. 6. The similarity error is acceptable, remaining within two decimal places with an initial value of 0.10 and stabilizing within three to four decimal places with an initial value of 0.30.

TABLE IV
AVERAGE COMPUTATIONAL AND COMMUNICATION COST OF EACH PHASE IN P3RS-MT

	<i>prop1</i>	<i>prop2</i>	<i>prop3</i>	<i>prop4</i>	<i>prop5</i>
<i>Comp.</i>	2.83s	0.02s	5.35s	0.003s	1.33s
<i>Comm.</i>	/	1.60MB	0.47MB	0.078KB	54KB

Efficiency: The evaluation focuses on the computational and communication costs of P3RS-MT, followed by an efficiency comparison.

We measured the average computational (*Comp.*) and communication (*Comm.*) costs for each phase in P3RS-MT, including **Key Generation** (*Prop1*), **Data Encryption and Uploading** (*Prop2*), **Similarity Computation** (*Prop3*), **Decryption** (*Prop4*), and **Personalized Recommendation Service** (*Prop5*), as shown in Table IV.

Key Generation is a one-time offline operation, and its impact on system performance is minimal.

The remaining four phases require both the user and server to stay online. In **Data Encryption and Uploading**, the computation and communication costs at the user side are relatively fixed at 19.02 ms and 1.60 MB, respectively. In **Similarity Computation**, the ciphertext similarity computation in MTF costs approximately 5.35 s, with a communication cost of approximately 0.47 MB. In **Decryption**, the computational cost at CSP is about 2.77 ms, with a communication cost of only 0.0078 KB. In **personalized recommendation service**, the decryption cost on the user side is approximately 1.3 s, with no communication cost, while the encryption on MTF is approximately 0.03 s, and incurs a communication cost of approximately 54 KB.

b) Comparison of accuracy and efficiency of P3RS-MT: In this subsection, we present a comprehensive performance analysis and comparison of P3RS-MT against existing schemes [15], [19], [25], [26], [27]. Before delving into the quantitative performance evaluation, we first establish a qualitative baseline by comparing the fundamental properties of these schemes across five critical dimensions: similarity detection technology, support for simultaneous homomorphic addition and multiplication, mechanisms for division transformation, support for real-number arithmetic, and IND-CPA security compliance. A detailed summary of this feature-based comparison is provided in Table V. Guided by the distinctions and limitations highlighted in Table V, we structure subsequent performance analysis into two specific comparative experiments.

First, as detailed in the Technology column of Table V, we compare the computational efficiency of our proposed WTC against other widely used similarity metrics that involve division operations: Cosine similarity [15], [25], Jaccard similarity [27], and the standard Tanimoto coefficient [19]. Accordingly, we select [15] as the representative for the Cosine comparison. Furthermore, to ensure a strictly fair comparison, all methods were re-implemented using CKKS scheme. This standardization is essential to support the real-number arithmetic required for division operations in each metric. To meet the multiplicative depth needed for two rounds of Newton iteration in CS-based method, we configure the modulus chain as

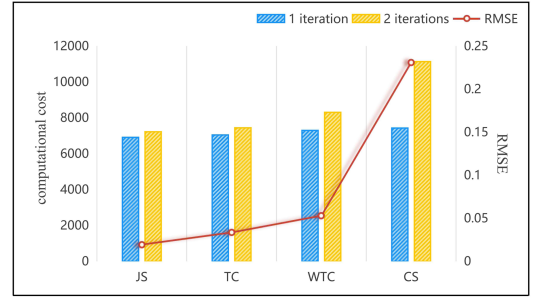


Fig. 7. Iteration computational cost and RMSE under Newton iteration of four methods.

[30, 25, 25, 25, 25, 25, 25, 25, 25, 25, 25, 25, 25, 30] with a scale of 2^{25} . The initial values for all methods are set to their optimal values. We then record the average computational cost of each method under one and two rounds of Newton iteration.

As shown in Fig. 7, JS-based [27] and TC-based [19] methods with two iterations incur less computational cost than CS-based method [15] with a single iteration. Although WTC used in P3RS-MT has a slightly higher cost than JS-based and TC-based methods, it still maintains a lower computational cost than CS-based method.

This is primarily because, beyond one iteration, CS-based method requires five ciphertext multiplications per iteration, compared to three for the other three methods. Furthermore, achieving higher prediction accuracy requires more iterations, causing the computational cost of CS-based method to increase significantly faster with increased iteration rounds.

Fig. 7 also shows the accuracy comparison of four RMSE-based methods over two iterations, averaged over ten experimental runs. RMSE is computed between P3RS-MT's similarity results e_i and those in plaintext r_i . The results indicate that JS-based [27], WTC-based and TC-based [19] methods maintain low computational errors, whereas CS-based method [15] yields a higher RMSE. This discrepancy is due to CS requiring more ciphertext multiplications at the same precision settings, leading to accumulating rounding errors and decreased accuracy in similarity computation.

Second, as detailed in the With division support column of Table V, we assess the computational overhead of the Newton iteration employed in P3RS-MT against other division transformation strategies used in [25], [26], [27]. Unlike the previous full-scheme simulation, this analysis focuses exclusively on the efficiency of the division operations, with the comparative results detailed in Table VI. Specifically, the bit-decomposition-based secure division in [26] incurs the highest computational burden, characterized by a complexity of $O(L \cdot T_{exp})$ and necessitating multiple rounds of interaction (3–5), which significantly creates communication bottlenecks. Similarly, [25] relies on a joint decryption protocol with a complexity of $O(\sqrt{V})$; this approach scales poorly as the data value range V increases. Although [27] achieves the lowest complexity via cross-multiplication, it functions strictly as a predicate protocol limited to pairwise comparisons, making it unsuitable for personalized recommendation scenarios that require the global ranking of

TABLE V
COMPARISON WITH EXISTING SCHEMES

Schemes	Application	Technology	Both HE-add and HE-mult support	With division support	With real number support	Security
SE-PPFM [19]	Fuzzy search	Tanimoto coefficient+ASPE	✓	Hadarnard Product+trapdoor	✓	Not IND-CPA
Vu-Thi et al. [25]	Recommendersystem	Cosine similarity+Elgamal+SMPC	✓	SMPC	×	IND-CPA
CryptoMask [15]	Facerecognition	Cosine similarity+BFV	✓	×	×	IND-CPA
Han et al. [26]	TravelRecommendation	Euclidean Distance+Paillier	×	Secure Divisionprotocol	×	IND-CPA
PPSKQ [27]	Range query	Jaccard similaity+Symmetric HE	✓	Cross-multiplication+Matrix Trace	×	IND-CPA
P3RS-MT	Medical tourism	Weighted Tanimoto coefficient+CKKS	✓	Newton iteration	✓	IND-CPA

TABLE VI
COMPARISON OF COMPUTATIONAL COST

schemes	round complexity (R)	computational cost
Vu-Thi et al. [25]	1	$O(T_{exp} + \sqrt{V} \cdot T_{mul})$
Han et al. [26]	3-5	$O(L \cdot T_{exp})$
PPSKQ [27]	1	$O(1) \cdot T_{marix_{mul}}$
P3RS-MT	0	$O(I \cdot T_{polymul})$

Note: T_{exp} : Modular exponentiation time, T_{mul} : Modular multiplication time, $T_{polymul}$: Polynomial multiplication time, $T_{marix_{mul}}$: Matrix multiplication time, V : Range of the plaintext value, I : Number of Newton iterations.

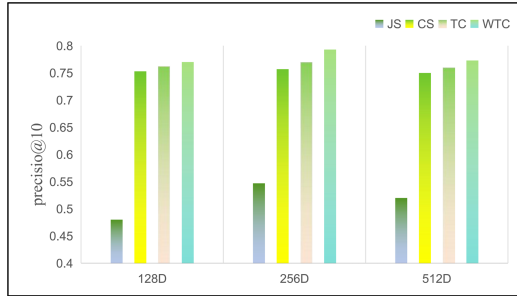


Fig. 8. Precision@K of four methods.

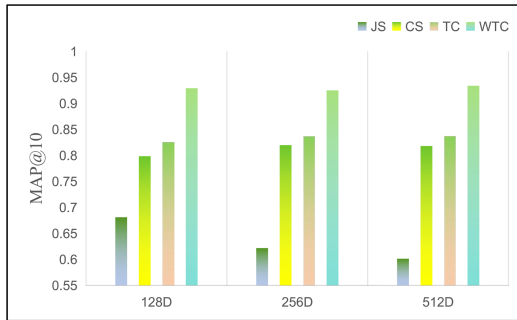


Fig. 9. MAP@K of four methods.

multiple candidates. In contrast, P3RS-MT employs Newton's method to approximate the inverse directly on ciphertext. This approach eliminates the need for interaction, thereby ensuring higher security while maintaining an acceptable computational overhead.

ii) Experiments on DARWIN: We conduct experiments on DARWIN to evaluate the accuracy of P3RS-MT with sparse medical data.

To satisfy the system's input requirements for sparse data, we applied Truncated SVD [42] to the raw DARWIN dataset. By calculating a low-rank approximation, the scheme maps high-dimensional sparse signals into a dense latent space (128, 256, or 512 dimensions). This process effectively suppresses inherent noise while significantly reducing the multiplicative

depth required for downstream encrypted similarity ranking. To simulate missing items in high-dimensional medical records, we applied a Top -K sparsification mechanism with a retention ratio of 10%. After sparsification, the active frequency f_j of each dimension across all samples is counted and normalized to the range $[0, 2]$, serving as the weight w_j .

Next, we use Precision@K and MAP@K to evaluate recommendation performance. In this experiment, we set $K=10$, selecting the top 10 most similar records for each query sample, with the number of query samples set to $N=30$. As shown in Figs. 8 and 9, P3RS-MT achieves strong performance in both Precision@K and MAP@K, with MAP@K consistently reaching 0.9—close to the ideal value of 1.

IX. CONCLUSION

This paper focuses on the privacy leakage of users' medical records during recommendation services in medical tourism, and proposes a privacy-preserving secure recommendation service, P3RS-MT. It enables service providers to effectively recommend the most suitable options for users without accessing their actual medical records. In P3RS-MT, we employ Newton iteration to convert division operations involved in WTC into multiplications, ensuring computational accuracy and efficiency. P3RS-MT has been proved to be secure against semi-honest service provider attacks. Extensive experiments on real datasets demonstrate that P3RS-MT achieves accurate recommendations while maintaining high efficiency for all participating parties.

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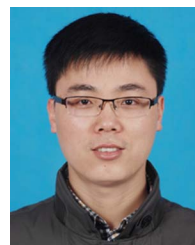
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